



Reasoning and learning with board game-based learning: A case study

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Abstract

Game-based learning (GBL) refers to the use of games to support learning in educational settings or as educational tools. The objective of this qualitative study is to gain an in-depth knowledge of what GBL means for the educational community of a public school in Spain and to explore the impact GBL has on the teaching-learning process. Data were collected through document analysis, observation during the game sessions (45 students), a focus group (6 teachers), and by in-depth interviews (1 family member, 1 teacher, and 2 students). Data analysis used constant comparison until category saturation was achieved. The participants conceptualized GBL around three main topics: (1) GBL characteristics, (2) participation, and (3) quality learning and reasoning. Thematic coding of their statements suggested that GBL encourages participation in school; this produces motivation and contributes to educational inclusion. GBL yields quality learning in both content and curricular competencies and favors the development of the executive functions (EFs), which are the basis of high-level thinking abilities. Results suggest that GBL is a favorable active methodology for the twenty-first-century school. The data is discussed in light of child development and learning in an educational context.

Keywords Game-based learning · Participation · Reasoning · Motivation · Executive functions · Metacognition · Cooperation

Quality education plays “a vital role in developing the knowledge, skills, attitudes, and values that enable people to contribute to and benefit from an inclusive and sustainable future” (OECD, Organisation for Economic Co-operation and Development, 2018, p.4). Schools are advised to implement methodologies that stimulate the development of twenty-first-century skills and motivate the participation of all students. Some of these skills are cognitive and metacognitive (e.g., critical and creative thinking, self-regulation, and learning to learn), and others are social and emotional (e.g., collaboration and cooperation, empathy, or self-efficacy) (Carr & Cameron-Rogers, 2016; OECD, 2018; Perkins, 2016). However, beyond twenty-first-century skills, the learning process is changing; in fact, it is a cultural phenomenon that can occur without books, teachers, and classrooms (Thomas & Seely-Brown,

2011). In this context, learning environments need to be spaces where “learners can explore, in a collective way, issues of interest and relevance, in ways that promote inquiry and questioning, through exploration, creativity and play” (Carr & Cameron-Rogers, 2016, p.11). Game-based learning (GBL) can play a unique role in facing the challenges of school today.

The Importance of Play

Games have been part of children’s lives in most cultures since the beginning of time. Classic authors in developmental psychology, such as Piaget and Vygotsky, have recognized the importance of playing in human learning and development (Plass et al., 2015). Play is a perfect learning experience and offers invaluable possibilities to think, learn, and develop knowledge (e.g., content), skills (e.g., cognitive, metacognitive, social, and emotional), and dispositions (e.g., attitudes and values, such as curiosity, and open-mindedness) (Salmon, 2015; Tishman et al., 1993).

Game-Based Learning

GBL is the use of games to encourage learning in didactic environments or as educational tools (Carr & Cameron-

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Rogers, 2016). There are different classifications of games, Table 1 shows the categories as defined by Carreras et al. (2010), Romero and Gómez (2016), and Victoria-Uribe et al. (2017).

Games are a metaphor for twenty-first-century learning because they contain the characteristics of the latest learning theories identified as appropriate for generating a learning context. Games are (1) active, (2) experiential, (3) contextualized, (4) social, (5) intrinsically motivated, and (6) problem-based (Carr & Cameron-Rogers, 2016; Plass et al., 2015; Viejo & Ortega-Ruiz, 2018).

Game-Based Learning: Motivation and Engagement

Games provide effective learning environments because they encourage motivation, engagement, adaptability, and graceful failure (Chen et al., 2015; Plass et al., 2015). Carr and Cameron-Rogers states that “game mechanics like badges, levels, and achievement boards provide students with goals that extrinsically motivate them to participate” (Carr & Cameron-Rogers, 2016, p.11). In contrast, other internal factors such as enjoyment or self-determination drive intrinsic motivation. Several studies suggested that GBL increases motivation and promotes effective learning, especially when complemented by collaborative learning (Chen et al., 2015; Chen & Law, 2016). In a game environment, the student has a feeling of control, accomplishment, and effectiveness over the game without fear of failure; they are just playing. Therefore, GBL methodology can benefit those students who do not feel committed or motivated by traditional learning methods (Carr & Cameron-Rogers, 2016; Eseryel et al., 2014).

Games can intrinsically motivate student engagement and facilitate a flow state (Csikszentmihalyi, 2000). This flow state favors students’ participation in mental activities that allows them to reach emotional satisfaction. Games permit the expression of different positive emotions: curiosity, joy, optimism, or pride and affect personal well-being, decision making, and the development of strategies (Lee & Hammer, 2011; Luna et al., 2018). As Huang (2011) affirms, “the playing process is fun, voluntary and intended to overcome challenges” (p. 694). Therefore, learning opportunities that games

provide motivate the effortless and autonomous feeling among students as they achieve the objectives or challenges established (Chen & Law, 2016). Along the same vein, feelings of competence, relatedness, interest, and self-efficacy are key elements in the process of maintaining people’s motivation; a playful methodology makes this possible (Eseryel et al., 2014; Ryan & Deci, 2000).

Games are flexible, and players can personalize or customize them. Games can engage each learner because they can adapt to different learners’ situations, knowledge levels, cognitive abilities, and emotions (Lee & Hammer, 2011; Plass et al., 2015). GBL allows failure, but in a unique way: failure can be graceful because gamers play time and again without serious consequences. Characteristics of game learning environments differ to such an extent from non-game environments (Plass et al., 2015).

Game-Based Learning: Executive Functions and Thinking Abilities

Several studies support that GBL can improve executive functions (EFs) when games are explicitly and rigorously designed for this purpose. DAVIS et al. (2015) provided computer training to children with ADHD. They found several EFs improved, such as inhibition, visual-spatial short-term memory, and working memory. However, these effects were specifically related to the type of game used in the intervention. Homer et al. (2018) asked high school students to play a video game designed to promote EFs for six weeks and found a significant improvement of their EFs. Oei and Patterson (2014) investigated the impact of playing different types of games on EFs and found that only the group that trained on physics-based puzzle games (non-action video games) improved several aspects of EF. Ramos et al. (2017) found that integrating children aged from 7 to 9 years in collecting games favored EF development. Results showed a strong prevalence of self-regulation and self-control in their interaction with others and an increase in the quality and adequacy of interpersonal relations.

Improving EFs is a continuous developmental process that is especially relevant in early childhood because the

Table 1 Different classifications on the types of games

Category	Types
Stages of cognitive development	Physical play, symbolic play, games with conventional rules, and building play.
Skills	Psychomotor, cognitive, social or affective games.
Material support	Games with material support: Board games and videogames. Games without material support: Games for developing motor skills
Social dimension	Competitive, cooperative and collaborative games.
Game mechanics	Collecting, capturing, building, dexterity or racing games.

maturation and structural development of new circuits and interconnections between important brain regions predominate during the first years. This process results in improved efficiency of cognitive-resource use between the ages of 10 and 12, resulting in significant performance differences among children and young people. Therefore, promoting the development of EFs in the first academic years is fundamental to enhancing children's development and learning (García et al., 2016). During the process of solving complex cognitive tasks, thinking skills can be improved by enhancing the use of EFs; however, selecting adequate tasks based on previous theoretical analysis is necessary. Therefore, each of the higher-level abilities to be instructed and the role of the EFs operating in them must be considered (García-Madruga et al., 2016).

Thinking and EFs have a significantly similar nature and dispositional components, and both reside in the prefrontal cortex. Therefore, EFs are essential for carrying out higher-order cognitive skills needed for effective academic performance, and games can be a useful tool for developing them (García-Madruga et al., 2016; Sanz de Acedo et al., 2012).

Game-Based Learning: Participation and Inclusion

Inclusive education promotes removing barriers that limit the presence, participation, and achievement of all learners (UNESCO, 2017). Inclusive education tries to facilitate quality education for all students and acknowledges diversity as something natural and intrinsic to all human beings (Velásquez et al., 2016).

The presence of students refers to equal opportunities to access education (UNESCO, 2017) and to the student's classroom presence at all times, as some schools often implement practices that segregate students (Booth & Ainscow, 2002). As Carr and Cameron-Rogers (2016) affirmed, GBL promotes learning outcomes in different ways and can facilitate students with different abilities sharing the game space. Thus, students who usually attend another class for educational reinforcement can, instead, remain in the general classroom with teachers using GBL methodology—achieving the first premise for educational inclusion (Flecha, 2015).

The second element considers participation. As an active methodology, GBL has a clear social component. It provides students with opportunities to participate in collaborative environments that encourage interacting with peers, developing an understanding of the other, learning to work interdependently, and resolving conflicts. All of these factors are key elements for educational inclusion (Hung et al., 2015; Luna et al., 2018). Participation in GBL occurs naturally because the game mechanics motivate the player, who must perform specific actions to achieve, in this case, a learning goal (Carr & Cameron-Rogers, 2016; D'Aprile et al., 2015).

The third element needed to advance toward inclusive education is achievement (Echeita & Ainscow, 2011). Achievement can be encouraged through active methodologies that favor student participation and allow each student—regardless of special educational needs—to construct their knowledge and prepare them for the future (Tavares & Sanches, 2013). GBL encourages all students to benefit from participating and learning by creating heterogeneously grouped play environments. In fact, extensive documentation shows that “low achievers benefit from the pace of instruction used for high-achieving pupils, demonstrating positive results in classroom arrangements that feature cooperative learning” (Flecha, 2015, p. 25). Thus, the learning process is supported when students participate and interact in GBL (Huang, 2011). According to Toppe et al. (2019), playing games produces a real-life context for children to experience different types of interaction, and this can be the origin of prosocial behaviors. Hanghøj et al. (2018) recommend using cooperative GBL because it increases the inclusion of students at risk and improves their social participation, motivation, and commitment to the curriculum.

In light of previously mentioned issues, GBL could be an appropriate methodology for students to develop knowledge, skills, attitudes, and values. Thus, this qualitative case study aims to explore the knowledge, opinions, and meanings that the education community (teachers, students, and family) attribute to the use of the GBL established in their school.

Method

This research employed a qualitative case study approach for an in-depth understanding of the education community's subjective and intersubjective reality around GBL and to give meaning to its classroom use (Baily et al., 2016). Following the conditions stated by Stake (2005) to shape the research design of a case study, our research is: (1) an exploratory case study because only one case has been analyzed; (2) holistic because investigates reality in an in-depth and balanced way, and offers a global vision of the phenomena studied and its complexity through different units of analysis; (3) intrinsic because has value in itself. We aimed to reach a better understanding through four data collection techniques: documentary analysis, a focus group, interviews, and observation (Stake, 2005). In short, this research's objective is to gain knowledge about GBL's specific and unique characteristics in the classroom context.

Participants

The sample in this study included teachers, students, and families selected using a purposive-sampling method. It is not a probability sample because its objective is to delve into

understanding the studied phenomenon and reach adequate category saturation (Hernández-Sampieri & Mendoza, 2018).

Participants in observations of the game sessions were 5th-grade ($n = 25$) and 6th-grade ($n = 20$) students. In Spanish educational system 5th and 6th graders are considered to be in Primary or Elementary School, children in this level are 11 and 12 years old. We considered their heterogeneous nature because some students had special educational needs (16% in the 5th grade and 16% in the 6th grade). The observation unit consisted of each round played by each group (4–5 players) during the observation sessions. A total of 24 observation sessions were held.

The focus group consisted of five form tutors: 40% from early childhood education ($n = 2$) and 60% from primary education ($n = 3$). This study considers form tutor as the teacher assigned to a class (Word Reference, 2020). The average age of the participating form tutors was 40.8 (D.T. = 2.56) and 60% were men. All of them had practiced GBL in their classrooms for a minimum of two years.

We conducted semi-structured interviews with students, family volunteers, and a teacher. We conducted three interviews with 5th- and 6th- grade students, with an average age of 11.45 (D.T. = 1.78 years old), and 33% were girls. Two of these participants were chosen because they were children with special education needs. Additionally, we interviewed one mother (41-years-old) who volunteered for motivating the board game sessions. Family volunteers play the role of invigorating and encouraging the board-game sessions. Family volunteers help and support the form tutor. We chose this participant because she regularly volunteered in game sessions and had two children with socioemotional difficulties attending the school. Finally, we interviewed the 5th-grade form tutor (42-years-old). We chose this participant because she had 4 years of experience with GBL, and 16% of her students had special educational needs. We determined that these interviews could provide many examples of the ways in which games encourage motivation and participation and contribute to the students' learning and reasoning.

Context

The participating school is in Castilla y León (Spain). The public school was opened in 2010 and is located in a new residential area. There were 220 students (121 boys and 99 girls) distributed among nine classrooms (three in early childhood education and six in primary education). One classroom was assigned for each course. The GBL project began during the 2016–17 academic year. At the time, the teachers received the training needed to carry out the GBL methodology in their classrooms. Board-game experts from Spain's Murcia University trained teachers in GBL. The methodology was progressively introduced during two academic years (2017–18 and 2018–19). To avoid problems with game distribution,

teachers allocated time, varying in duration from 30 min to one hour per session, and space for GBL during the week. Family participation was considered when designing the GBL sessions. The form tutor assigned a family volunteer to oversee a board game and motivate each group of students. Meanwhile, the form tutor lent support to each of the groups and observed how the game was functioning. The school had 45 board games that allowed for changing the games every trimester. Even if it was not the goal of this study, the detailed participant and context descriptions make it possible to transfer the research's essence to similar contexts (Hernández-Sampieri & Mendoza, 2018). All activities, games and procedures involved in the study were conducted in Spanish.

Data Collection

The techniques used to gather information followed a typical case study design: document analysis (DA), focus group (FG), semi-structured interview (I), and observation (O) (Álvarez & San Fabián, 2012). We analyzed school documents (Education Project, Curricular Project, and the Annual General Program) to observe the antecedents and implementation that the school granted to active methodologies, especially GBL (Hernández-Sampieri & Mendoza, 2018).

We established a focus group to analyze the interaction among teachers who implemented GBL and how they constructed meaning (knowledge, attitudes, and opinions) about this innovative experience (Hernández-Sampieri & Mendoza, 2018). We (all the researchers) prepared and revised a script in advance (see Table 2). First, we asked about the positive and negative aspects of GBL to the participants. Then, with minimum intervention, researchers noted in a control list whether the topics belonging to the object of the study appeared to encourage (1) inclusion and students' participation; (2) family participation; (3) students' learning results; (4) motivation; (5) cognitive skills, such as reasoning or reflection.

Table 2 Script used to invigorate the focus group with teachers. Source: Own elaboration

Focus group
Impact of GBL among students.
- Positive aspects of GBL
- Negative aspects of GBL
Delve into or induce the following topics if they do not appear:
- Have they talked about inclusion and participation?
- Have they talked about how family participation affects students?
- Have they talked about how they believe it affects the students' results?
- Have they talked about students' motivation?
- Have they talked about the cognitive abilities that favor GBL, such as planning, strategic thinking, reflection, etc., that can favor students' reasoning/thinking?

Moreover, we conducted semi-structured interviews with teachers, students, and families to understand their internal and in-depth perspectives about GBL implementation in their school. Interviews included opinions, expressions of feelings, and questions about knowledge (Hernández-Sampieri & Mendoza, 2018). These interviews enabled us to obtain first-hand accounts that showed the unique significance that different educational agents have with GBL, mainly with how this methodology affects students with special needs (Mertens, 2010). The script of these interviews was previously defined with questions adapted to students, teachers, and families (see Table 3). During conversations with participants, we added new questions if a topic of interest appeared.

We observed different GBL sessions to understand the GBL process, participants' interactions during play, and the learning they can achieve. In qualitative research, the investigator, during the observation, must be in touch with the reality studied in order to understand the intersubjectivity developed. In this case, the observers, two researchers from this study, had moderate participation, taking part exceptionally during the game sessions observation and recording them for an in-depth analysis of what happened (Hernández-Sampieri & Mendoza, 2018). The information collected by multiple data sources allowed us to use the triangulation process to test the validity, credibility, and confirmability through the convergence of data from different sources (Guba, 1989; Hernández-Sampieri & Mendoza, 2018).

Procedure

First, we contacted the educational authorities (School Management Team and Provincial Management) to obtain the necessary permits to access the center and collect data. We performed all procedures per the ethical standards of the institutional and national research committees and the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This study was approved by Regional Institutional Review Boards. With regard to ethical issues, we obtained informed consent from the participants in the study. Because the students were minors, their parents or legal guardians signed the consent. Once we compiled the informed consents, we began collecting the data following the chronology in Table 4.

We analyzed the following school documents to observe the presence and treatment of active methodologies, especially GBL: (1) the Education Project, which establishes the framework of the center's government and autonomy, its following of the guidelines, and intervention strategies included in the management project; (2) the Curricular Project, which defines the center's methodological and pedagogical approaches developed in the Didactic Programs of each course; and (3) the Annual General Program, which establishes the aspects related to the center's organization and operation during a school year (Junta de Castilla y León, 2016).

We devoted our first days in the school to collecting observational data to understand, in a contextualized way, how the

Table 3 Script for semi-structured interviews with teachers, students and families. Source: Own elaboration

Interview: teachers	Interview: students	Interview: families
1. Do you have students with difficulties in your class? What characteristics do they have and what type of support do they need?	1. What do you like most about school? Can you order them from the one you like the most to the one you like the least? If s/he doesn't mention games, ask him/her where s/he would place them.	1. How do you value your participation in the board game sessions carried out in your son's/daughter's class?
2. How do you think GBL is affecting those students who have special needs? And how does it affect students in general? Positive and negative aspects of GBL.	2. Why do you like games?	2. What is the use of attending these sessions as a volunteer?
3. How do you think GBL favors the participation of students with and without special needs?	3. How do you feel when you play board games in the class? Why do you think you feel like this?	3. How do you think the GBL sessions are affecting your son/daughter? Positive and negative aspects.
4. How do you think GBL favors the academic results of students with and without special needs?	4. What do you think you learn when you play board games?	4. Do you think your son/daughter learns contents or abilities with board games? Which ones?
5. How do you think GBL favors motivation among students with and without special needs?	5. What is the use of all the things you have learned?	4. How does your son/daughter participate? 5. Do you think games have helped him/her in any way in other spheres apart from games? How?
6. Do you think GBL favors cognitive abilities such as planning, reflection or reasoning?	6. Delve into or induce the following topics, if they haven't appeared previously: cooperation, participation, inclusion, motivation, examples of games that use specific cognitive abilities or curricular contents.	6. Do you think games motivate your child? How?

Table 4 Chronology of the technique for collecting data

Technique for collecting data	Participants
1. Document analysis (DA)	Education Project, Curricular Project, Annual General Program.
2. Observation (O)	Students: 5th grade (n=25)
3. Focus group (FG)	Teachers: Early Education (n=2) and Primary Education (n=3)
4. Observation (O)	Students: 6th grade (n=20)
5. Interviews (I)	Student 1 (6th grade)
	Student 2 (6th grade)
	Student 3 (6th grade)
	Parent 1 (6th grade)
	Teacher 1 (5th grade)

board game sessions were conducted and their possible implications on an educational level. The observations were conducted according to the following script: (1) audiovisual recording of the game sessions; (2) qualitative note-taking for referential adequacy; (3) analysis of the recording to extract the categories. All the researchers analyzed the recordings to implement effective triangulation in the data analysis.

After observing the board game sessions, we observed the focus group with five teachers in their classrooms. The observation lasted 45 min with no incidents; a high agreement appeared with every topic addressed. One of the researchers acted as a moderator, so he could obtain details about the participants' experiences and reasoning to improve the discussion (Hennink, 2014). The shared reflections were recorded and transcribed. The participants confirmed the transcription met their approval before we analyzed it. The high degree of accord among participants and the transcription confirmation denote that our results reflected the reality of the classroom with respect to the GBL methodology. This provided the study with confirmability and validity (Guba, 1989).

After analyzing the focus group, we continued observations to confirm that the emerging categories reflected the GBL sessions (Baily et al., 2016).

We proceeded with semi-structured interviews with students, families, and teachers. The transcriptions were confirmed by those interviewed to achieve the confirmability criterion (Guba, 1989). The interviews with the students, volunteer mother, and the form tutor were conducted in the classroom, with the aim of participants answering in a natural context. The interviews lasted between 5 and 20 min each.

To ensure this research's credibility, we collected referential adequacy data. We gathered this additional material using field notes to document the context during data collection (Guba, 1989), which included informal conversations with students, teachers, and families who volunteered in the game sessions. As Hernández-Sampieri and Mendoza (2018) indicated, the referential adequacy data allowed us to "record all the dimensions of the events or experiences" (p. 456). In this way, we not only considered what the participants told us (verbal communication) but also how they told us (non-verbal and paraverbal communication).

Data Analysis

To analyze the data, we followed an inductive process with three cyclical stages (Hernández-Sampieri & Mendoza, 2018). During the first phase of discovery and data collection, we identified categories and topics after analyzing the field notes, the focus group transcriptions, the interviews, and the literature consulted. In the second phase, we encoded the topics and categories to begin organizing the data. The initial categories selected from the bibliographic analysis were checked throughout the research process so that categories not appearing were eliminated, and those emerging from the data were added (Guba, 1989; Hernández-Sampieri & Mendoza, 2018). The third phase was to develop the data interpretation system, considering the context where it was collected. We outlined a visual representation model in this phase, which reflects the relationships between categories and topics that emerged from the data of the different units of analysis. We used the computer software ATLAS.ti version 8 in the three phases described. The video with the recorded game session observations, the focus group transcriptions, and the interviews were analyzed by two researchers to avoid personal bias and to gain different perspectives, which provided stability to the study. The fact that different investigators gathered similar data in the field of research and conducted the same analysis should provide similar results, as shown in this case (Hernández-Sampieri & Mendoza, 2018). The calculations of Cohen's Kappa parameter to measure the inter-rater reliability was 0.92, which indicated very high agreement. Finally, we considered the following quality standards from qualitative research: dependency, credibility, transferability, and confirmability (Guba, 1989), as we have mentioned throughout the report.

Results

This research explored the knowledge, attitudes, opinions, and meanings that the application of GBL holds in the educational community (teachers, students, and family) through document

analysis, observations, focus groups, and semi-structured interviews.

Following an inductive sequence, a concept map has been created with different levels to reproduce the data and present the findings in a meaningful way (see Figure 4). The relative presence of each category in the total number of encoded fragments has been estimated to visualize each category's presence, taking into consideration all the units of analysis. Table 5 facilitates the visual identification of themes and categories to construct the final conclusions gradually (Daley, 2004).

Educational Characteristics of GBL

Educational community responded GBL favors motivation, positive emotions, and that it is a flexible methodology that can be adapted to the classroom's diversity (see Fig. 1).

About 6% of the assertions referred to motivation. Motivation is one of the most noteworthy aspects of literature referencing GBL (Chen & Law, 2016; Lee & Hammer, 2011; Plass et al., 2015). Games favor both intrinsic and extrinsic motivation, which allows students to experience a flow state that makes them push themselves to participate and learn (Csikszentmihalyi, 2000).

"Play produces the need to reason in a specific way to achieve a goal. Then, if you do not create a need, if you do not favor motivation in the kid to do something—an effort—the student will not do it. The game creates this need and motivates to do it." (FG, *teacher4*)

"Yes, absolutely true, they motivate a lot. One of the students with special educational needs is totally passionate about the game moments." (I, *teacher*)

"The game! It motivates my children indeed. It triggers them to make friends, meet classmates..." (I, *mother*)

"What I like best of school is break time and games." (I, *student1*)

A total of 4% of the comments collected referred to positive emotions. Positive emotions that students feel when they are playing are clearly related to motivation. Educational community states that play provides pleasure, fun, and confidence.

"Both the children with special educational needs and the rest of students have an optimal emotional state, so they feel good, calm, and safe with the things they are doing because they feel no one is assessing them; they are just playing." (I, *teacher*)

"We always see them more confident when they play than when they do a task." (FG, *teacher2*)

"One of my kids, very shy, participates in the game without any problem; he likes board games. He interacts

with his peers, and he even smiles and laughs at jokes."

(I, *mother*)

"Games help me learn to see boring things in a funny way." (I, *student 2*)

Finally, 2.5% of the assertions referred to the flexibility of games that allow all students to engage. Games have dynamics or established rules that can be modified to facilitate all students' participation. This quality can address knowledge levels and experience that appear naturally in the classroom (Carr & Cameron-Rogers, 2016). The flexibility to meet any child affects educational inclusion positively.

"How does it benefit them? It benefits them because the game includes different paces and learning styles. Then, we do not see here if a student has a specific level or not. We notice whether the student understands the rules, the game structure, etc. and this vision equals all the students." (I, *teacher*)

"Students change the rules because they do not understand them; the game dynamics is improved with the changes introduced by the pupils." (O, *Observer2*)

"On the other hand, I also highlight the fact that, as there are such different games, there is always a game in which those students who have more problems can do it properly." (I, *teacher*)

GBL and Quality Learning

According to the data, GBL provides quality learnings that helped students improve their results and progress. Figure 2 shows information about the educational community's opinion on the quality of learning in GBL. This concept is complex because it ranges from the perceived connection between GBL and content learning to the connection between GBL and high-level thinking abilities. A total of 16.55% of the comments coincided with the idea that GBL increases a quality learning.

GBL is considered a methodology that could promote curricular content learning (e.g., oral and written expression) and basic competences for the twenty-first century (e.g., linguistic, mathematical, or learning to learn). The school documents (Education Project, Curricular Project, and Didactic Program) revised in the document review, justify the use of GBL through the development of (1) curricular content, (2) common transversal elements, and (3) key competencies. Teachers, families, and students consider GBL a methodology that could favor content learning.

"Use mental agility in mental arithmetic exercises and to study. But you learn more; you remember more and have a better oral expression." (I, *student1*)

Table 5 Categories, theoretical definition and frequency derived from the data analysis

Topic	Categories	
Educational characteristics of GBL (16.5%)		
Motivation (6%)	Response produced by an incentive that directs and maintains the person’s behavior towards goals or specific purposes. Intrinsic motivation focuses on the personal abilities, and extrinsic motivation in external rewards (Eseryel et al., 2014).	
Positive emotions (4%)	Those that produce pleasant sensations, that derive from a win-win situation for the individual due to this reason (Ryan & Deci, 2000).	
Adaptability or flexibility (2.5%)	It is the characteristic board games have to adapt to any player, no matter what their previous knowledge and circumstances are (Carr & Cameron-Rogers, 2016).	
Flow state (0.75%)	Mental state in which a person performing an activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment in the process of the activity. There is intrinsic motivation, because the skills and abilities we have allow us to meet the challenge, and the challenge itself produces extrinsic motivation (Csikszentmihalyi, 2000).	
Quality learning (16.5%)		
Curricular learning (8.25%)	Key competences (5.25%)	Integrated learning that combines knowledge, skills and attitudes (Booth & Ainscow, 2002). Some of the competences considered by the Spanish educational system are social competence, civic competence, learning to learn, sense of initiative and entrepreneurship.
	Contents (2%)	Knowledge, skills or attitudes students have to acquire during the teaching-learning process (Booth & Ainscow, 2002).
Cognitive abilities (7%)	Executive Functions (8%)	Cognitive and metacognitive skills that direct our behaviour towards the realization of a goal (Homer et al., 2018).
	Reasoning (4.25%)	Set of cognitive processes that consist of connecting ideas according to some specific rules, in order to solve a problem or create a product (García-Madruga et al., 2016).
	Creativity (1.25%)	Ability to invent or create something (Viejo & Ortega-Ruiz, 2018).
Participation and inclusion (7.5%)		
Students	Student Participation (6.5%)	Possibility given to students to take part in the teaching-learning process (Flecha, 2015).
	Cooperation (3.5%)	Collaborative actions developed by a group of people to achieve a common goal (Booth & Ainscow, 2002).
	Identity (1.25%)	Games become part of the students’ tastes and interest (Carr & Rogers, 2016).
Family	Family participation (6.25%)	Possibility that families are given to intervene in the teaching-learning process of their children from within the classroom (Flecha, 2015).
	Connection with the school (3.75%)	Families maintain a closer contact with the school (García et al., 2016).
	Coordination with teachers (1%)	Family-teachers coordination is closer and more organic (García et al., 2016).
Teachers	Learning strategies (0.3%)	As families take part in the teaching-learning process that happens in the classroom, they learn strategies that can be used in their houses (García et al., 2016).
	Attention to diversity (3%)	An open methodology allows the teacher to pay a better attention to the specific needs’ students have, especially of those with special educational needs (Soriano, et al., 2017).

"I think GBL develops more meaningful and long-lasting learning. Games exercise executive functions, which are basic. The use of these processes can affect positively the academic progress of students with difficulties." (I, *teacher*)

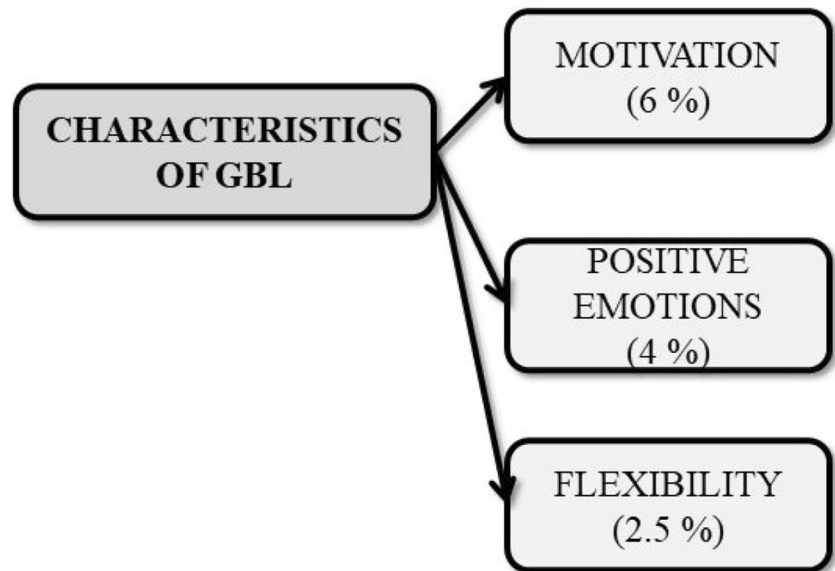
"Because we learn; for example, with *Picopico*, we learn addition, work in teams in aspects that are not so academic, like language" (I, *student3*)

"There is a game with rabbits that have to jump in order to get a carrot. It has to estimate and calculate measures

to jump and to add planks so that it can advance. Then, if they put the needed measure, they can advance. If not, they won't go further. So, you are working measures, calculus, and estimation; it is, no doubt, an academic objective." (FG, *teacher1*)

The issue of the game's curricular integration appeared in different units of analysis. Teachers were motivated and interested in continuing to incorporate this learning tool and achieving a higher prevalence of games in the teaching and learning process.

Fig. 1 Emerging categories that constitute the theme characteristics of GBL. Source: Own elaboration



The difficulty of assessing learning acquired from GBL arose as a derived topic when this methodology is introduced.

“We must be able to integrate games even more in the classroom: in every subject and every day. We use

games to improve oral and written expression in language, and we should use other games in natural science and other areas.” (I, *teacher*)

“Using games as a focus of interest, as a daily tool in every subject.” (I, *teacher*)

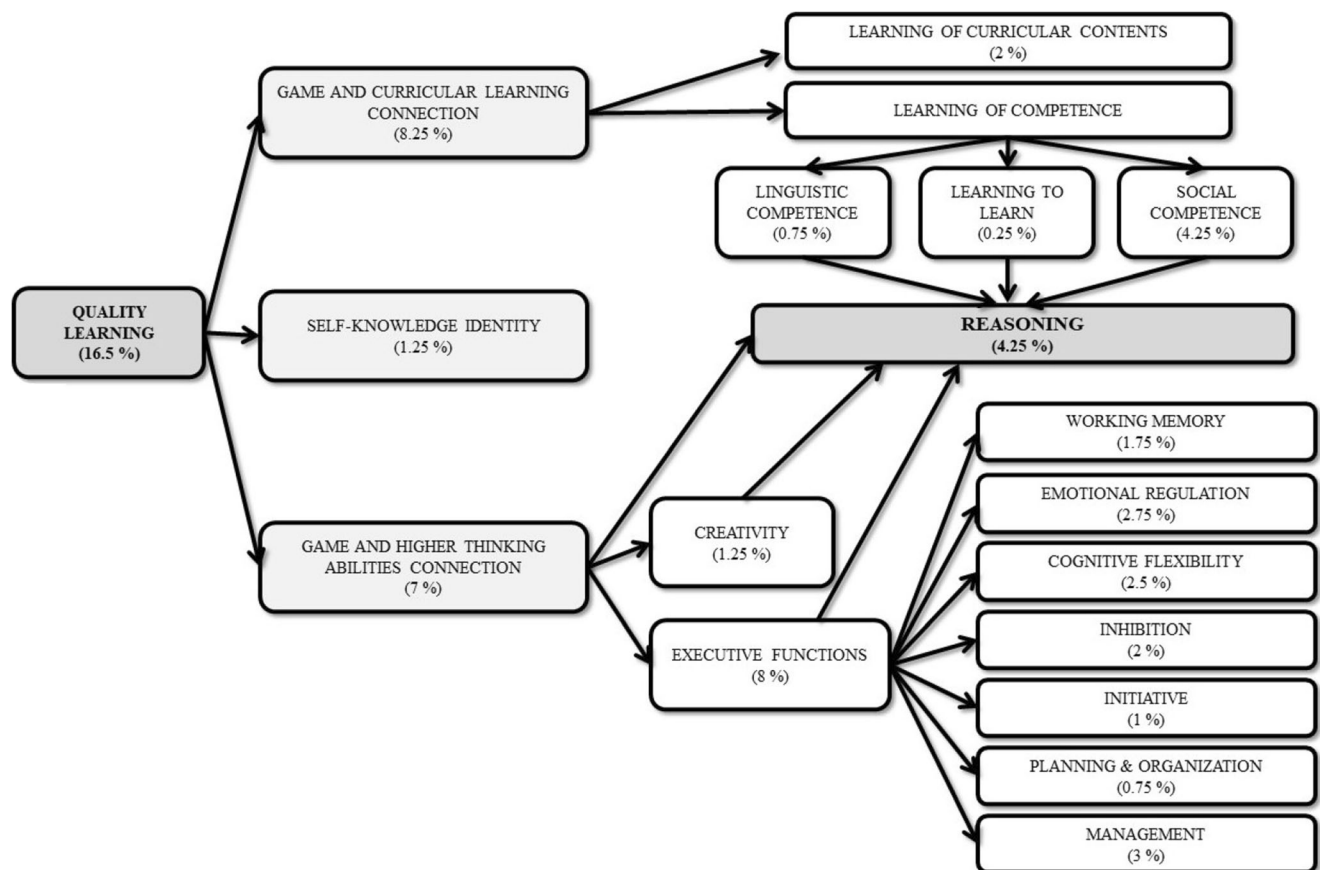


Fig. 2 Emerging categories that constitutes the theme Quality Learning. Source: Own elaboration

Nevertheless, teachers found some problems fulfilling this objective due to the educational system's structure (subjects, timetables) and to difficulties introducing GBL in their evaluation.

"It seems clear that we are training different abilities that will be useful to improve their curricular competence. This is what matters, actually. However, measuring it is very difficult. I mean, you provide students with memory, attention, and abilities that help them improve indirectly at a curricular level. But it is very difficult to evaluate whether they have a better curricular competence when they play." (FG, *teacher3*)

"Another problem is assessment. When we evaluate, we assess the achievement of some contents. We do not pay attention to whether the student's strategies have improved; we could prepare a tool to assess his strategy." (FG, *teacher1*)

"I am sure that it affects learning. Another question I have as a teacher is how could I evaluate it, how to quantify it? I have asked myself it, but I do not know how to do it." (FG, *teacher2*)

Regarding learning competencies, social competence acquisition has been remarkable; GBL could improve aspects such as the students' social abilities, cooperation, or socioemotional self-regulation.

"Taking part in the games has let me know the way children behave, how they interact, whether there is competition among them when they play, if order and turn-taking is maintained, whether they get on well with each other and cooperate. In a nutshell, it is useful to observe their social relationships." (I, *mother*)

"Relationships among us affect games because if you get on well with somebody, you don't want to attack the virus, but if you don't like the other, you are going to attack him with everything." (I, *student3*)

"There are two negative aspects that can be transformed into positive. First, some children feel frustrated with some games, so we can help them have a low frustration tolerance. We can also help them transform the negative aspects of competitiveness and leadership into positive facets. In short, negative aspects are possibilities that will enrich the person." (FG, *teacher2*)

GBL seems to promote high-level cognitive abilities such as creativity and reasoning and EF in students, which are essential to developing these abilities. Many studies have supported the relationship between GBL and the improvement of reasoning and EF (Buelow et al., 2015; DAVIS et al., 2015; Homer et al., 2018; Oei & Patterson, 2014; Ramos et al., 2017).

According to the data, GBL mainly promotes the following EF: (1) management, (2) emotional regulation, (3) cognitive flexibility, (4) inhibition, (5) working memory, (6) initiative, and (7) planning and organization. In the study, 8.25% of the assertions referred to the link between GBL and curricular learning, whereas 7% referred to the connection between GBL and acquiring high-order cognitive abilities. Other categories emerging as subjects from extensive comments were reasoning (4.5%) and EF (8%).

"GBL allows to improve abilities like perception, memory, attention, strategy, and others that are difficult to work with in other ways." (FG, *teacher3*)

"Something you get with board games and not with technologies is that you are less lazy. I mean, your brain will not retain information when you use the new technologies because you know it is going to be there. However, when playing a board game, you have to push yourself to think, to make a choice, to draw up strategies... You take pains to solve it or do it. This will help you afterwards, when you grow older to solve problems, like interacting in your job." (I, *mother*)

"Teamwork, memory, and rapidity." (I, *student3*)

Finally, we would like to mention another type of learning that emerged from the analyzed data: GBL helps students increase their self-awareness as players, which enlarges their self-knowledge and defines their identity. A total of 1.5% of the assertions referred to this aspect:

"I find it interesting the way that, depending on the game, some students use more- or less-complex strategies. This can be related to the fact that games can adapt to any educational situation or player. As a consequence, all students can benefit by increasing their self-esteem, because all can succeed." (O, *Observer2*)

GBL: Participation and Inclusion

GBL can favor students' participation because they are motivated to take part in the game, and the only way to acquire a quality learning experience is by participating (Huang, 2011; Tavares & Sanches, 2013). GBL allows families' participation too, because permit them to go to school and get involved in the educational activity. Because the activity is a game, families seem to feel more motivated and capable of implementing it with the teachers and the students. School document analysis identified family and students' participation in their objectives:

- Encourage family participation as a key element to guarantee greater inclusion.

- Favor students' inclusion in the school and social life, paying special attention to students with special education needs
- Encourage the use of new methodologies in the teaching-learning process, such as: Cooperative Learning, Project-Based Learning, Game-Based Learning, and Visible Thinking (DA, Education Project, p. 3).

About 17.75% of the comments originating from all the techniques and participants referred to participation. Figure 3 summarizes the information regarding how the educational community members consider that GBL favors participation.

"I think they are less ashamed of taking part in a game than when speaking in front of the class. Some introverted students let themselves go during the games. It is easier interacting in groups of four, as in the game, than with all the class." (FG, *teacher1*)

"In the specific case of one of my pupils, I can hardly understand him when he speaks in the assembly. However, he participates in the games and interacts better. Play is children's natural condition, especially with four years old kids." (FG, *teacher2*)

Different studies have confirmed the relationship between students' participation and the quality of learning (Carr & Cameron-Rogers, 2016; D'Aprile et al., 2015; Flecha, 2015; Tavares & Sanches, 2013). In this study, 6.5% of the comments referred to this aspect and the way GBL, indirectly facilitates cooperation among students, making higher-level learning possible (Plass, 2015).

"As most of the games are collaborative and not individualistic, we have to be a team; we help each other,

and so all the differences among us disappear. We all learn." (I, *teacher*)

"It is true that we emphasize working as a team, being together, and solving things with the others." (FG, *teacher5*)

Family participation could improve family-school connection and promotes educational inclusion. An adequate connection between family and school increases coordination, and this aspect encourages a more appropriate and individualized treatment of each student. This improvement occurs because the interventions of both educational agents are more coherent and follow the same course of action. Moreover, family participation teaches parents learning strategies utilized in the school so that they can use them at home (Flecha, 2015; García et al., 2016 ; Valls-Carol et al., 2014). A total of 5% of the comments referred to the family-school connection and its impact on the quality of learning.

"Parents also see what these games provide. As a teacher, you explain them what you are working with each game, and they see that students are also having fun." (I, *teacher*)

"Volunteer families say before entering the game session that it is an interesting project and that they learn a lot when they come to school to participate." (O, *Observer1*)

An adequate connection between families and school encouraged a more suitable and individualized treatment for each student because both educational agents followed the same line, and their actions were more coherent (García et al.,

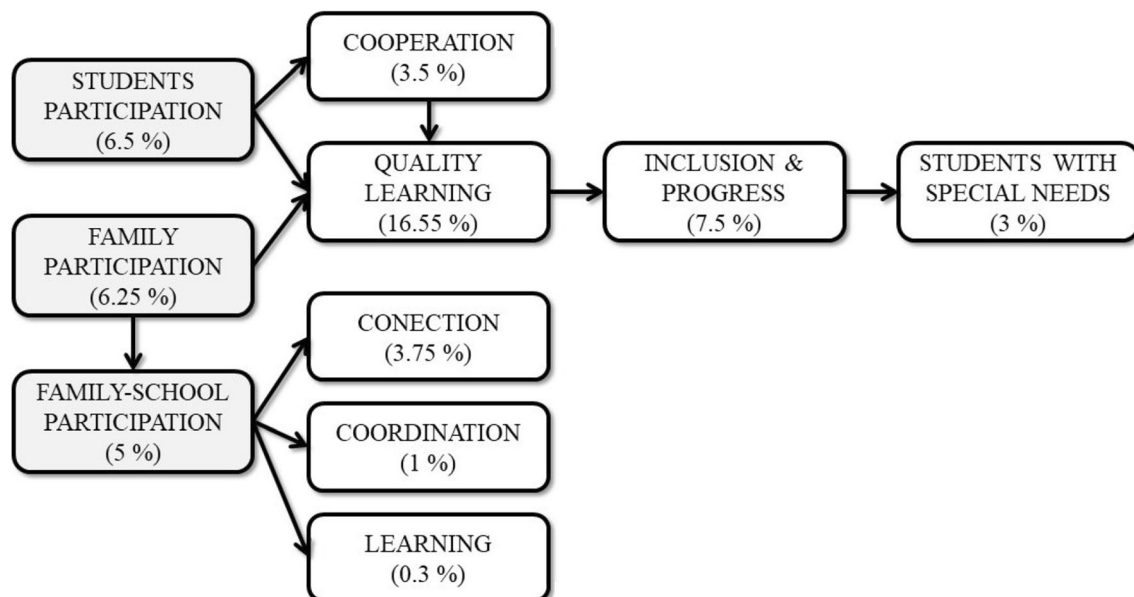


Fig. 3 Emerging categories that constitutes the theme participation. Source: Own elaboration

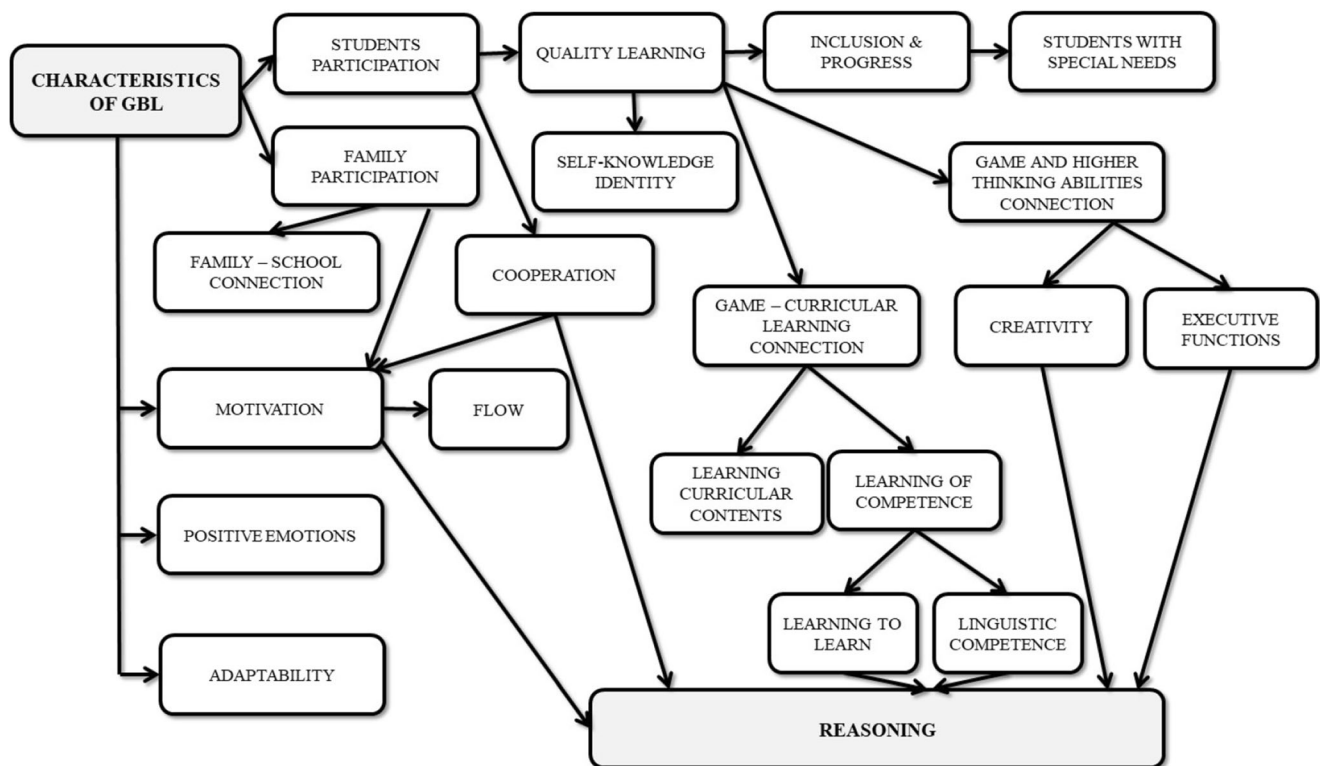


Fig. 4 final concept map with all themes and categories emerged and their relationships. Source: Own elaboration

2016; Valls-Carol et al., 2014). Moreover, the family's presence in the classroom could motivate students.

"I have fun most of the times I play because I am with my classmates. Mothers also get involved, and I like it."
(I, student 1)

Once the insecurity barrier breaks, teachers confirm that family participation in the classroom could be positive, and it benefits their teaching because families learn educational strategies used in the school and can apply them at home. Additionally, they can understand and appreciate the teacher's work (Flecha, 2015; García et al., 2016; Valls-Carol et al., 2014).

"Family participation has been one of the most difficult aspects for me because it seems that we sometimes want to lock ourselves up in the classroom. But it is the opposite. Now I see that if you open yourself and let others see, things go better, and you have less problems." (FG, teacher4)

"We have even extrapolated games, that is, they do not remain in the classroom: families know them and have bought them. Families are persuaded of the benefits of games." (FG, teacher2)

"The fact that parents can be in the classroom is very positive because they not only experience the classroom but also see the problems we have and can understand many things." (FG, teacher3)

To sum up, this section presented the results of the different themes (GBL educational characteristics, quality learning, and participation) and their connection with the different categories, as portrayed in a final concept map that reflects the conclusions of this study (see Fig. 4).

One of the focus group's participating teachers affirmed this closing point:

"If I reflect on the way a child faces a game, bearing in mind what we have mentioned about motivation, curiosity, pleasure, etc. and the way the student faces other tasks that are usually more repetitive, it is so different! I think all the emotion attached to solving things through games is richer and more creative, so the child is going to develop better, from the neuronal view." (FG, teacher2)

Conclusion

This case study aims to explore the knowledge, opinions, and meanings that the education community (teachers, students, and family) have regarding GBL implementation in the school. Findings suggest that GBL could be an adequate methodology to encourage the twenty-first-century key competencies as reasoning, problem solving, creativity, or social competence among students. Furthermore, results suggest that

GBL could be a methodology that favors educational inclusion, as it encourages student and family participation and promotes learning for all students.

Educational community agreed that educational characteristics of games could favor motivation, positive emotions and flexibility. Games seem to provide effective learning environments because they encourage extrinsic and intrinsic motivation, pleasure and joy, engagement, and participation and interaction among all the students. In recent years, several such approaches have been suggested (e.g. Carr & Cameron-Rogers, 2016; Chen et al., 2015; Chen & Law, 2016; Eseryel et al., 2014). Human being needs to feel competent and autonomous in order to be intrinsically motivated. Games could fulfill these needs because they provide stimulating, challenging, but safe learning environments. Failure during play is not perceived as incompetence to the same extent as in other academic environments. In addition, games are played in environments of interaction and relationship with other players (e.g. peers, teachers, families) so they would meet the three elements of intrinsic motivation (Ryan & Deci, 2000). Therefore, GBL could help students actively participate in building their knowledge and developing high-level cognitive abilities.

Regarding the cognitive abilities that can be developed through GBL, educational community highlighted EFs, especially working memory, emotional regulation, cognitive flexibility, planning and organization, inhibition, initiative, and management. These results are in line with another research conducted by Dovis et al. (2015), Homer et al. (2018), Oei and Patterson (2014), and Ramos et al. (2017) and their implications are important because EFs are the basis of high-level thinking abilities (understanding, reasoning, and solving problems). The above-mentioned EFs are classified within the metacognitive executive functions, and Ardila (2018) stated they are the neuropsychological basis of what is defined as general intelligence. High-level thinking abilities should be promoted because childhood and GBL could play an important role in educational culture. Educational policies should support teacher training in GBL because this methodology could improve student EFs, which are necessary for future academic and professional success. In fact, Hanghøj et al. (2018) stated that the teacher's role in GBL is a key factor in facilitating achievement and acquiring benefits from the game, as GBL's connection to the educational curriculum is fundamental. Integrating GBL is not only about play but also playing specific games in a specific way, with clear objectives connected to the curriculum.

Educational community suggests that GBL could be a valuable tool to address classroom diversity—understanding diversity in its broad sense, as a natural and intrinsic characteristic of our students, and more specifically when we refer to students with special educational needs. GBL could be considered an inclusive methodology because of the inherent

flexibility of games. First, it encourages conditions for all students to remain in the classroom, regardless of educational needs or a different academic profile. Second, it promotes participation among peers, with teachers, or volunteer families. Student and family participation are two of the educational measures that researchers attribute to educational success (UNESCO, 2017).

Our study's observations reveal that the ease of supervising board games encourages families to participate more frequently in class activities. This participation opens doors to improving the educational institution in collaboration with all the agents who intervene in school: students, teachers, and families. Family participation seems to facilitate individualized attention for students, and it can be a motivating element because students can verify that families are interested in their schoolwork. Moreover, as results show, participation could benefit teachers because when families help in the classroom, teachers are free to move around and observe students, getting to know them better. As the participating teachers stated, this autonomy allows them to adjust the teaching-learning process, adapt it to each student, and increase the quality of the educational process. Furthermore, research confirms that teachers feel more supported when they see families get involved in classroom activities and support teachers' initiatives. According to the results of this study, the family benefits from learning educational strategies that can be applied at home. This connection allows for more cohesiveness between the two environments where children spend most of their time: school and home.

Considering competences, we note the data shows that games could facilitate real-learning situations where social competencies are essentially developed. This learning could favor all participating students and encourage their learning of cognitive processes that are essential in everyday life, such as EFs. Developing adequate social competence in all the students, regardless of whether they have special educational needs or not, can only be acquired in social contexts, through social interaction. The results of this study suggest that GBL could facilitate this type of learning, such as frustration tolerance, emotional control, and respect for peers in a social environment that is real, natural, and familiar to the student. If we consider students with social difficulties, this aspect becomes of interest, and we must emphasize its importance. As observed in the analyzed context, the game could provide a playful and fun environment where students with difficulties can feel safe, display their social abilities with confidence, and overcome shyness or the fear of being tested. Consequently, learning will be more significant and more meaningful in a fun and relaxed environment.

In short, GBL is not only an active methodology that facilitates knowledge acquisition (e.g., game-curricular learning connections). It could also develop skills (e.g., social, cognitive, and metacognitive), attitudes, and values (e.g.,

cooperation and collaboration) for future-ready students. It could also help mobilize these aspects into practical and educational contexts. Thus, GBL is compatible with the learning framework that the OECD considers for 2030 (OECD, 2018). As Piaget and Vygotsky each affirmed, play is a natural and vital activity for children's development and learning, so it is imperative to consider its educational potential inside and outside schools (Plass et al., 2015).

Due to the case study methodology used, we cannot generalize and transfer the results obtained in the research to other contexts or situations. Still, we can produce new research questions that will find answers with different methodologies, either qualitative or quantitative. The map of categories presented in this study can be the starting point for generating new studies and checking whether they can be replicated in other contexts. It can be analyzed in a longitudinal way to determine the impact this methodology has on students or if the results can be generalized. The case study's subjectivity is usually seen as both a benefit and a limitation in the literature (Álvarez & San Fabián, 2012). In this study, we have tried to access the benefits of being a researcher-participant, as one of the researchers is also a teacher in the school we studied. We have also attempted to minimize the researcher's impact on the study's participants, as they are familiar with him.

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Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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